Endogenizing Nature in Production Theory: A Framework of Economic Commensalism

Fernando Souza da Vieira[1] Alexandre Chibebe Nicolella[2]

[1]Faculty of Economics, Administration and Accounting of Ribeirão Preto, University of São Paulo, Brazil

Abstract

The relationship between economic activity and the natural environment has long been marked by a persistent tension between productive expansion and ecological preservation.

Problem

Keywords: economic commensalism; environmental economics; rational incentives; commons; carbon markets.

1. Introduction

The relationship between economic activity and the natural environment has long been marked by a persistent tension between productive expansion and ecological preservation. The environment is essential to human welfare and economic production, yet the depletion of natural resources and the degradation of ecological systems can undermine the very productive capacity that economic activity seeks to expand. Conversely, production, investment, and technological progress generate employment, income, and improvements in living standards, so environmental policy cannot be evaluated independently of its effects on private incentives and productive viability. The relevant economic problem is therefore not whether production or preservation should prevail, but under what conditions the use of nature can remain compatible with the ecological functions on which present and future production depend.

Economic theory does not ignore nature, on the contrary. Land, natural resources, environmental costs, externalities, and common-pool resources have been extensively examined across different strands of the literature, such as Pigou (2010), Coase (1960), Hardin (1968), Ostrom (1990), and Dasgupta (2001), as we will see in detail in Section 3.

[1] fernandodvieira@usp.br
[2] nicolella@usp.br

In canonical representations of production, such as the Cobb-Douglas function, output is usually modeled as a function of capital and labor, sometimes with land or natural resources appearing as inputs, but ecological systems are not generally represented as dynamic cofactors that affect future productivity, strategic incentives, and intertemporal welfare. Instead, the environment often appears as a stock, a constraint, a source of externalities, or an object of corrective regulation, and this representation has economic consequences.

When ecological functionality remains outside the private decision problem, preservation tends to appear as a legal obligation, a compliance cost, or an opportunity cost imposed on production. Its benefits are frequently delayed, uncertain, spatially diffuse, or collectively appropriated, whereas the returns from resource conversion may be immediate, measurable, and privately captured. Even an agent who recognizes the long-term importance of ecological integrity may therefore rationally choose short-term expansion when the expected private return from degradation exceeds the expected private return from preservation. The divergence is intensified by credit constraints, competitive pressures, incomplete information, weak enforcement, and uncertainty regarding future policies.

Governamental command-and-control regulation and penalties, carbon markets, environmental certification, payments for ecosystem services, and fiscal incentives translate parts of these insights into policy. These approaches may happen incour in some improvements, but they address different components of the problem and do not automatically establish a unified relationship among ecological conditions, production, private payoffs, and territorially grounded governance to the point of balancing private and collective interests efficiently, both for the economy and for the environmental use of the resources involved.

Nevertheless, the main remaining gap is neither the absence of nature from economic thought nor the absence of instruments — whether legal, financial, institutional, or even moral — intended to protect it. The key issue is that the environment, often treated as exogenous, should be integrated into a theoretical framework in which nature, production, incentives, and governance are simultaneously represented and analytically connected, so as to clarify the conditions under which preservation and productive viability can be efficiently reconciled.

Against this background, the article asks: *under what conditions can ecological preservation be represented and institutionally organized as an endogenous and incentive-compatible component of production decisions, so that private productive viability remains consistent with the maintenance of ecological functionality?*

This question replaces a simple opposition between use and preservation with a

problem of institutional and productive design. It also shifts the analysis from whether economic activity causes any environmental impact to whether it systematically erodes the ecological conditions required for future production and social welfare.

This work therefore seeks to integrate relationships that are often treated separately in the literature, and to derive their joint implications for production theory, private incentives, and environmental governance.

The following hypotheses guide this article: (H1): endogeneidade; (H2) governança local; (H3) racionalidade intertemporal.

Existing approaches illuminate parts of these relationships, but their joint theoretical implications for production, incentives, and environmental governance remain insufficiently developed.

As we will argue, the spatial dimension is especially important. Carbon compensation programs may be fungible for the purpose of accounting for a global atmospheric stock, but equivalent quantities of carbon do not imply equivalence in biodiversity, hydrological regulation, soil resilience, or ecological connectivity. Compensation undertaken elsewhere may satisfy a market or regulatory requirement without restoring the local ecological functions that sustain a particular productive territory. Likewise, adding an environmental variable E to a production function does not by itself make the environment endogenous: endogeneity requires that E affect productive outcomes while its condition also responds, directly or indirectly, to production, preservation, institutional rules, and ecological feedback.

The article aims to make three principal contributions. First, to propose a more precise representation of nature in production theory by distinguishing the mere inclusion of a natural-resource input from the endogeneity of an ecological state that affects productivity and responds to economic decisions. Second, to connect this representation to private incentives and ecologically grounded governance, emphasizing that environmental equivalence at the level of carbon accounting does not necessarily preserve locally specific ecosystem functions. Third, it introduces economic commensalism as a testable analytical framework and derives theoretical conditions for distinguishing productive use that maintains ecological functionality from resource-use regimes that erode the basis of future production. The proposed framework does not reject regulation, enforcement, markets, or collective governance; it reorganizes their roles around the problem of making preservation a more robust component of the agent's rational decision rather than an exclusively external burden.

Methodologically, the article adopts a theoretical-analytical approach that combines production theory, environmental economics, and institutional analysis. It first

reconstructs how externalities, property rights, common-pool resources, and natural capital illuminate distinct parts of the production–preservation problem. It then develops a preliminary model in which the environmental state enters the production structure and the payoff of productive agents, while incentives and governance affect the relative return to preservation and exploitation. The model is used to compare alternative ecological–economic strategies and to derive the conditions under which a commensal configuration can emerge as an individually rational and socially desirable outcome. The analysis remains deliberately foundational: it establishes the conceptual and static analytical basis upon which intertemporal dynamics, uncertainty, learning, and empirical testing can subsequently be developed.

The remainder of the article is organized as follows. Section 2 contextualizes the production–preservation problem and the limitations of current environmental governance. Section 3 reviews the relevant literature, moving from externalities and property rights to common-pool governance and natural capital, and identifies the elements that remain analytically disconnected. Section 4 develops the conceptual and analytical model of economic commensalism, defines the alternative ecological–economic regimes, and states the relevant equilibrium conditions. Section 5 presents the theoretical results and comparative implications of the model. Section 6 discusses their relationship to the existing literature, their institutional meaning, and the limitations of the proposed framework. Section 7 concludes and indicates directions for dynamic formalization and empirical assessment.

2. Problem Context

—> Mecanismos atuais de governança ambiental (regulação, precificação, mercados de carbono, etc.) são insuficientes para lidar com a complexidade e a dinâmica dos sistemas socioecológicos.

—> Dados e evidências mostram que, apesar dos esforços, as tendências de degradação ambiental e perda de biodiversidade continuam, indicando falhas estruturais nos modelos de governança atuais.

—> Citar esforços internacionais (Acordo de Paris, Objetivos de Desenvolvimento Sustentável) e nacionais (políticas ambientais, leis de proteção) que não têm conseguido reverter as tendências negativas.

—> Usar dados de desmatamento, perda de biodiversidade, emissões de gases de efeito estufa, Brasil e mundo, para ilustrar a insuficiência dos mecanismos atuais.

–> Concluir que é necessário repensar os fundamentos teóricos e práticos da governança ambiental, buscando abordagens que integrem melhor a natureza como parte dos sistemas econômicos e sociais, e que sejam mais adaptativas, inclusivas e eficazes.

3. Literature Review

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Posicionar explicitamente em relação a Dasgupta (2021): O relatório The Economics of Biodiversity (Dasgupta Review) endogeneiza a natureza de forma sofisticada. Mostrar no que seu framework vai além desse trabalho já canônico.

4. Conceptual and Analytical Framework

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5. Theoretical Results

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6. Discussion

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7. Final Remarks

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References

COASE, R. H. The problem of social cost. **Journal of Law and Economics**, v. 3, p. 1–44, 1960.

DASGUPTA, P. **Human Well-Being and the Natural Environment**. 1st. ed. Oxford: Oxford University Press, 2001. ISBN 9780199247882.

HARDIN, G. The tragedy of the commons. **Science**, v. 162, n. 3859, p. 1243–1248, 1968.

OSTROM, E. **Governing the Commons: The Evolution of Institutions for Collective Action**. Cambridge: Cambridge University Press, 1990.

PIGOU, A. C. The online library of liberty. **Online Library of Liberty**, 2010.